

Chalaka Samith

COMPUTER ENGINEERING UNDERGRADUATE

☎ +94 77 4299094 ✉ chalakasamith@gmail.com 🌐 chalaka.online 📄 github.com/chala2001 in Chalaka Samith

About Me

Computer Engineering undergraduate (GPA 3.971/4.00, First Class) and an SRE intern at WSO2. I take a system the whole way: designing it, building the full stack, then deploying and running it on the cloud and on Kubernetes. Most recently, AI agents that take real work off the team.

Education

University of Peradeniya

BSc.Eng(Hons.) in Computer Engineering, GPA: 3.971/4.00, First Class

Peradeniya, Sri Lanka

2022 - Present

Royal College

G.C.E. Advanced Level, Physical Science, 3As | Z-score: 2.0291

Colombo 07, Sri Lanka

2018 - 2020

Experience

WSO2

Site Reliability Engineering Intern, On-site

Colombo, Sri Lanka

Apr 2026 - Present

1. **Compliance Evidence Submission Portal** | 📄 Repo

Agentic Browser Automation | In Production

- Replaced the team's **manual evidence collection** with an AI agent that drives cloud consoles and **GitHub** in a real browser, captures what each compliance control needs as a **screenshot or PDF**, and files it automatically.
- Added **task templates** so one instruction covers every resource the agent finds, instead of one per resource. A page export that already carries a link skips the AI entirely, dropping the cost of the most common job to **nothing**.
- Wrote the **threat model** and defended it in security review, which took the portal **into production** as a WSO2 internal application. The same work closed an open route where anyone guessing a file's ID could download evidence without signing in, now behind links that **expire in fifteen minutes**.
- Packaged the agent as an **installable command-line tool** and wrote the repository's first **release pipeline**: a version tag builds the wheel and publishes a GitHub Release, so an engineer installs it in **one command**, with first-party actions only so no unreviewed code ever holds the repository's tokens.
- Built the **FastAPI** backend and deployed it on **WSO2 Choreo**, storing data in **Azure Database for PostgreSQL** and evidence files in **Azure Blob Storage**, behind **enterprise SSO** on every request, with a check that stops the agent's own token from gaining admin rights. Covered by **410 tests** that run against a real database rather than mocks.

Tech: Python, FastAPI, Azure PostgreSQL, Azure Blob Storage, browser-use, Playwright, Azure OpenAI, React, TypeScript, WSO2 Asgardeo, OAuth 2.0, WSO2 Choreo, GitHub Actions, pytest

2. **SRE Ops Pilot (SREP), Grounded Assistant for the SRE Team**

RAG & LLM Evaluation

- Stopped the assistant **inventing answers** by checking its work in code instead of trusting it: the backend opens the document behind every citation and confirms the quoted line is really there, hiding any source that fails and warning the reader.
- Skipped the usual vector database and fed **whole documents** instead, because cutting them into pieces separates a table's numbers from the headings that give them meaning. Better document matching then brought the questions needing the whole library down from **44 of 65 to 21**, and the average request from 53k to 40k tokens.
- Wrote a **65-question test set**, 15 of them deliberately unanswerable, and chose the model on results rather than reputation: the winner scored **63 of 65**, correctly refused **all 25 trick questions**, and answered in 5.1 seconds instead of 12.2.
- Added **enterprise SSO**, saved conversations, per-person cost tracking, an **admin audit log** and a spend cap, then cut **three database round trips to one** by reading a user's identity from their sign-in token instead of writing first. Now at **552 backend and 115 frontend tests**.
- Wrote the **threat model** and trust-boundary document mapping all **nine interactions**, now in WSO2's security review: the last gate before it goes live.

Tech: Python, FastAPI, Azure OpenAI, PostgreSQL, Alembic, SSE Streaming, React, TypeScript, react-query, WSO2 Asgardeo, OAuth 2.0, pytest, Vitest

University of Peradeniya

Undergraduate Teaching Assistant, Part-time

Peradeniya, Sri Lanka

Sep 2023 - Mar 2026

CO322 Data Structures & Algorithms | CO225 Software Construction | CO2030 Data Structures & Algorithms II

Projects

1. **CloudCare HMS, AWS Well-Architected Infrastructure** | 📄 Repo

Personal | Cloud, IaC

- Built a production-style AWS environment **entirely in code**: **nine independent Terraform stacks** with shared, locked state, a private database spread across two availability zones, and layered firewall rules in front of it.
- Deployed through **GitHub Actions using short-lived credentials**, so no permanent AWS key exists anywhere, and ran it with a monitoring dashboard and **eight alarms** over a self-healing server group.

Tech: AWS, Terraform, GitHub Actions, OIDC, EC2, ALB, RDS, DynamoDB, Lambda, CloudFront, Secrets Manager, CloudWatch, Docker

2. **CloudCare HMS, EKS Microservices & Observability** |  Repo *Personal* | *Kubernetes, SRE*
- Split one large application into **four smaller services** on **Kubernetes**, each packaged with **Helm** and given its own deploy pipeline, so shipping one no longer redeploy the others.
 - Cut scale-up time from **five minutes to thirty seconds**, made rollback a **single command** by tagging every build with its commit, and watched the cluster through **Prometheus, Grafana and Loki**.
- Tech:** AWS EKS, Kubernetes, Helm, Terraform, Prometheus, Grafana, Loki, ALB Ingress, ECR, GitHub Actions, FastAPI, Docker
3. **LMS University, Full-Stack Digital Education Platform** |  Repo *Personal* | *Full-Stack*
- Built one **Spring Boot** API serving two clients at once, a **React** website and a **React Native** mobile app: a social feed, live messaging, an admin dashboard, and a job board where alumni hire students.
 - Moved it from Docker Compose onto **Kubernetes**, keeping the database's data intact across restarts and its passwords out of the configuration files.
- Tech:** Spring Boot, React, React Native (Expo), MySQL, Docker, Kubernetes, JWT, WebSockets, NGINX
4. **Explainable AI Driven Zero-Trust Anomaly Detection for Encrypted Traffic** *Final Year Research* | *iPURSE 2026*
- Designed a system that finds attacks inside **encrypted network traffic without ever decrypting it**, so privacy stays intact while threats are still caught.
 - Paired it with **Explainable AI** so an analyst sees *why* a connection was flagged rather than only that it was, and measured it against the **CIC-IDS-2017** benchmark.
 - Presented and defended the work before an **external expert panel** at **iPURSE 2026**, the Peradeniya University International Research Symposium, where the **abstract** was accepted into the conference proceedings.
- Tech:** Python, Semi-Supervised ML, Explainable AI, CIC-IDS-2017, NumPy, Matplotlib
5. **Lightify, Smart Home IoT Lighting System** |  Repo |  Web *Group* | *My scope: firmware*
- Wrote the **ESP32 firmware**, in C and C++, that the whole system runs on: reading the occupancy sensors, driving the lights, and letting a new device join over Wi-Fi or Bluetooth without being reprogrammed.
 - Connected the devices to **AWS IoT Core** so the backend could read sensor data and send commands back in real time. Presented at **EngEx 2025**.
- Tech:** ESP32, C, C++, AWS IoT Core, MQTT, Wi-Fi and Bluetooth Provisioning, Spring Boot, MongoDB
6. **Artery Resolver, Cardiovascular Risk Analysis** |  Repo *Group, with Faculty of Medicine* | *My scope: app, data, CV model*
- Automated **Flow-Mediated Dilation**, a test for early cardiovascular risk that clinicians were measuring by hand: the tool reads an **ultrasound video**, detects the artery walls with **OpenCV**, and tracks how far the artery widens as blood flows through it.
 - Paired that with **blood-flow velocity** analysis to turn each scan into a **summarised report** a doctor can act on, and built the **Java** desktop app and **MySQL** database behind it, with every record scoped so a doctor reaches only their own patients.
- Tech:** Java, Java Swing, JDBC, MySQL, Python, OpenCV, Computer Vision, Ultrasound Video Processing

Technical Skills

- Languages:** Python, Java, TypeScript, JavaScript, C, C++, SQL, Bash
- Backend & Web:** FastAPI, Spring Boot, Node.js, React, React Native (Expo), REST, WebSockets, SSE
- Cloud & IaC:** AWS, Azure, WSO2 Choreo, Terraform, Helm, Docker, Kubernetes (EKS), Linux, NGINX
- CI/CD & Observability:** GitHub Actions, OIDC Federation, Jenkins, Prometheus, Grafana, Loki, CloudWatch
- Data & Messaging:** PostgreSQL, MySQL, MongoDB, Redis, SQLAlchemy, Alembic, Azure Blob, Kafka, MQTT
- AI & Agents:** Azure OpenAI, Claude API, RAG, LLM Eval Harnesses, browser-use, Playwright, LangChain, OpenCV
- Security & Tools:** Threat Modeling, Enterprise SSO, WSO2 Asgardeo, OAuth 2.0, OIDC, JWT, Git, Claude Code, pytest

Professional Development

- Terraform, **Top 10% Globally** (TestDome, 2026)
- Introduction to Kubernetes LFS158 (Linux Foundation, 2026)
- AWS Cloud Practitioner Essentials (Amazon Web Services, 2026)

Extra-Curricular Activities

- EngEx 2025:** presented *Lightify* at the University of Peradeniya 75th Anniversary Engineering Exhibition
- Student Member**, IESL (2025 - Present) | **Member**, ACES, UoP (2023 - Present) | **Member**, AIESEC (2022 - 2024)
- Maths & Science Tutor**, Arunella Education Welfare Programme, student-led seminars for rural schools Jul - Dec 2023

References

Dr. Suneth Namal Karunarathna

Senior Lecturer, Dept. of Computer Engineering
University of Peradeniya, Sri Lanka | namal@eng.pdn.ac.lk

Mr. M.D. Rovin Shanila

Associate Technical Lead, WSO2 (Pvt) Ltd
Colombo 04, Sri Lanka | rovinshanila@gmail.com